UNIVERSITY OF LISBON

ISEG- Lisbon School of Economics and Management

Normal Season Exam - January 2024

Advanced Econometrics

Module Convenor:
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Duration: TWO HOURS

Use answer booklets of ISEG (folhas de teste do ISEG)

Instructions (please read before starting): Write in a clear legible manner in ink/ballpoint. Do not use pencils or erasable pens. Approved calculators are permitted. Only one sheet (2 sides A4) of notes made exclusively by the student may be consulted (no material distributed by the teacher in any form is allowed). Whenever conducting a test use a 5% significance level unless stated otherwise. Also be sure to state null and alternative hypotheses, null distribution (with degrees of freedom), rejection criterion (critical values and rejection region) and outcome. If you are asked to derive something, give all intermediate steps also. Do not answer questions with a "yes" or "no" only, but carefully justify your answer.

Section A - Topics in Microeconometrics

Question 1

Consider a random variable y and a scalar regressor x. The density function of y conditional of on the regressor x is given by

$$f(y|x) = \lambda^2 y \exp(-\lambda y), y > 0,$$

and $\lambda = 2\exp(-x\beta_0)$, where β_0 is a scalar parameter. Note that $E(y|x) = \exp(x\beta_0)$ and $Var(y|x) = [\exp(x\beta_0)]^2 / 2$. Our objective is to estimate the true parameter β_0. Suppose that a random sample $\{(y_i, x_i)'\}_{i=1}^n$ is available and that the regressor x is exogenous.

(a) *(2 Marks)* Write-down the log-likelihood function of the problem.

(b) *(1 Mark)* The score-vector is given by

$$S(\beta) = \sum_{i=1}^n s_i(\beta), \text{ where } s_i(\beta) = 2 \times \frac{y_i - \exp(x_i\beta)}{\exp(x_i\beta)} x_i,$$

Show that $E(S(\beta_0)) = 0$.

(c) *(2 Marks)* Let $\hat{\beta}$ be the maximum likelihood estimator of β_0. Obtain the asymptotic variance of $\sqrt{n}\left(\hat{\beta} - \beta_0\right)$, assuming that the density function is correctly specified.

(d) *(2 Marks)* Prove that the information matrix identity holds in this model.

Question 2

(2.5 Marks) Consider the following latent variable model

$$y^* = x'\beta_0 + u,$$

where y^* is not observed, x is a vector of exogenous regressors, β_0 is a vector of parameters and $u \sim \mathcal{N}(0, 1)$, that is the cumulative distribution function of u is given by $\Phi(\cdot)$. Assume that x and u are independent. Furthermore, suppose that the researcher observes

$$y = \begin{cases} 0 & \text{if } y^* \leq 0 \\ 1 & \text{if } y^* > 0 \end{cases}.$$

It is suspected that there is a probability τ of a value of $y^* \leq 0$ being reported as being strictly larger than zero. That is, the data may be "one-inflated". Assuming that a sample of n independent observations is available and the regressors are exogenous. write down the log-likelihood function for this problem.

Section B - Topics in Time Series

Question 3

Consider the $VAR(1)$ model $z_t = \Phi_1 z_{t-1} + \varepsilon_t$ where $z_t = (z_{1t}, z_{2t})'$ and ε_t is a 2×1 vector of white noise processes with

$$var(\varepsilon_t) = \begin{bmatrix} \sigma_1^2 & \sigma_{12} \\ \sigma_{12} & \sigma_2^2 \end{bmatrix}.$$

(a) *(2 marks)* Let

$$\Phi_1 = \begin{bmatrix} 1.1 & -0.3 \\ 0.6 & 0.2 \end{bmatrix}.$$

Obtain the roots of the characteristic equation and show that the process is stationary.

(b) *(2 marks)* Obtain the values of the elements of the matrices Ψ_ℓ, for $\ell = 0, 1, 2$ in the infinite moving average representation $z_t = \sum_{\ell=0}^{\infty} \Psi_\ell \varepsilon_{t-\ell}$.

(c) *(1 marks)* Obtain the impulse response function for z_{2t} to a shock to the variable z_{1t} of size $\sigma_1 = 1$, for horizons $\ell = 0, 1, 2$.

Question 4

In this question we investigate the relationship between wages and employment using data from the Current Employment Statistics (CES) project of the U.S. Bureau of Labour Statistics (BLS). For simplicity we focus on the data of one particular industry: Men's & boys' furnishings (standard industrial classification 232). We consider the following variables on this industry:
lemp - natural logarithm of employment in thousands;
lwage - hourly wages in dollars;
lcpi - natural logarithm of the consumer Price Index, 1982-1984 = 100.

(a) *(1.5 Marks)* Consider the information provided in the following table on the ADF and Phillips-Perron tests (intercept, no time trend).

	ADF		Phillips-Perron
	test statistic	lag-length	test statistic
lemp	0.7134	15	0.2044
lwage	0.0924	12	−0.0130
lcpi	1.4965	2	0.7493

Are the variables *lemp*, *lwage* and *lcpi* stationary?

(b) *(1 Marks)* The observed value of the Engle-Granger statistic to test for cointegration between *lemp* and *lwages* is 0.401435 (intercept, no time trend, lag-lenght 15).Are *lemp* and *lwages* cointegrated?

(c) *(1 Mark)* Which are the limitations of Engle-Granger test for cointegration?

(d) *(1 Mark)* Briefly outline the Johansen's methodology for testing cointegration between a set of variables in the context of a VAR.

(e) *(1 Mark)* Consider the results of the Johansen cointegration test *lemp*, *lwage* and *lcpi* (with lag length 14)

Johansen tests of H_0 : rank= r vs. H_1 : rank> r
(No deterministic trend)

r	Trace statistic	5% Critical Value	1% Critical Value
0	45.78937	34.91	41.07
1	18.77199	19.96	24.60
2	6.611721	9.24	12.97

What do you conclude about the number of cointegrating relations at 5% level and 1% level?

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